low-salt mixture

1kg → 0.02

x kg → $0.02x$ kg of salt

high-salt mixture

1kg → 0.96

y kg → 0.96y kg of salt

final mixture

1kg → 0.04

24 kg →$(0.04)(24)$kg of salt

x kg of low-salt mixture (2% salt)

y kg of high-salt mixture (96% salt)

Final mixture $= 24$ kg (4% salt)

equation?

$2\% = \frac{2}{100} = 0.02$

$96\% = \frac{96}{100} =$

0.96

$4\% = \frac{4}{100} = 0.04$

Salt from the **low-salt mixture** $= 0.02x$

Salt from the **high-salt mixture**$= 0.96y$

Total salt in the **final mixture** $= (0.04)(24)$

Salt from the low-salt mixture + Salt from the high-salt mixture = Total salt

$0.02x + 0.96y = (0.04)(24)$

$0.02x + 0.96y = (0.04)(24)$